THE LEGENDRE'S EQUATION

$$(1-x)^2 y'' - 2xy' + \alpha(\alpha+1)y = 0$$

- Second order, Linear, ODE
- α is a real number, $\alpha > -1$
- Extensive applications:
  - Numerical Integration formula (Gaussian Quadrature)
  - Boundary value problems
    (Steady-state temperature within a solid ball when temperature at points on the boundary are known.)



Legendre's equation: it looks intimidating until you realize it's a just a polynomial factory with a fancy entrance.

Your professor is sticking to his teaching philosophy...
"Same distination. Many ways to get there."



METHOD 1: For when the exam clock is ticking!

METHOD 2: For when you want to understand what just happened.

Solution to a special Legendre's equation where, $\alpha = 1$:

SHORT METHOD: Answer first

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

since $n = 1$

$$(1-x^2)y'' - 2xy' + 2y = 0 \Rightarrow y'' - x^2y'' - 2xy' + 2y = 0$$

STEP-1 $\left[y(x) = \sum_{n=0}^{\infty} c_n x^n \right]$ Power series of $y(x)$

STEP-2 Termwise differentiation & addition

We know $y(x) = c_0 + c_1 x + c_2 x^2 + \cdots$

$$2y = 2c_0 + 2c_1 x + 2c_2 x^2 + 2c_3 x^3 + 2c_4 x^4 + \cdots$$

$$-2xy' = \quad -2c_1 x - 4c_2 x^2 - 6c_3 x^3 - 8c_4 x^4 + \cdots$$

$$-x^2 y'' = \quad\quad -2c_2 x^2 - 6c_3 x^3 - 12c_4 x^4 + \cdots$$

$$y'' = 2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \cdots$$

Adding $x^0, x^1, x^2, x^3, x^4, \ldots$ terms & equate to zero

x^0: $2c_0 + 2c_2 = 0 \Rightarrow \boxed{c_2 = -c_0}$

x^1: $2c_1 - 2c_1 + 6c_3 = 0 \Rightarrow \boxed{c_3 = 0}$

x^2: $2c_2 - 4c_2 - 2c_2 + 12c_4 = 0 \Rightarrow -4c_2 = -12c_4$

$\Rightarrow +4c_0 = -12c_4$

x^3: $+2c_3 - 6c_3 - 6c_3 + 20c_5 = 0 \Rightarrow \boxed{c_4 = -1/3}$

$\Rightarrow -10c_3 + 20c_5 = 0 \Rightarrow \boxed{c_5 = 0}$ ($c_3 = 0$)

x^4: $2c_4 - 8c_4 - 12c_4 + 30c_6 = 0 \Rightarrow -18c_4 = -30c_6$

$-18/-3 = -30c_6$

$\boxed{c_6 = -1/5}$

$$\therefore\ y(x) = C_1 x + C_0\left(1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 - \cdots\right)$$

C_0, C_1 are arbitrary constants

$P_2(x) = P_1(x) = x$

$-Q_2(x) = -Q_1(x) = 1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 - \cdots$

[Two solutions] [special case where $n = 1$]

Two solutions are members of a family of functions called "Legendre polynomials"

KVB

LONG METHOD: For answer & understanding

POWER SERIES SOLUTION OF LEGENDRE'S EQUATION:

$$(1-x^2)y'' - 2xy' + 2y = 0$$

We solve: $(1-x^2)y'' - 2xy' + 2y = 0$

This is Legendre's Equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

with $n = 1$, since $n(n+1) = 2$

SOLUTION:

STEP-1 Write the second-order linear ODE in a standard form: $y'' + P(x)y' + Q(x)y = 0$

Defn. [ORDINARY POINT]	A point x_0 is an ordinary point if both $P(x)$ and $Q(x)$ are analytic at x_0, meaning each has a convergent Taylor series about x_0.
[SINGULAR POINT]	If either $P(x)$ or $Q(x)$ fails to be analytic at x_0, x_0 is a singular point.

Divide ODE by the leading coefficient $(1-x^2)$ —

$$y'' - \left(\frac{2x}{1-x^2}\right)y' + \left(\frac{2}{1-x^2}\right)y = 0$$

compare with $y'' + P(x)y' + Q(x)y = 0$

we have $\left[P(x) = \frac{-2x}{1-x^2}\right]$, $\left[Q(x) = \frac{2}{1-x^2}\right]$

STEP-2 Check analyticity of $P(x)$ & $Q(x)$ at $x=0$:

Both $P(x)$ & $Q(x)$ are rational functions. A rational function is analytic everywhere its denominator is non-zero.

At $x=0$, $1-x^2 = 1 \neq 0$

$\therefore$ $P(x)$ & $Q(x)$ are analytic at $x=0$

$\therefore$ $\boxed{x=0}$ → is an ordinary point,

STEP-3 Locate singular points & calculate radius of convergence:

The denominator of $P(x)$ & $Q(x)$ vanishes when $1-x^2 = 0$

$\therefore$ $\boxed{x = \pm 1}$ → are singular points

(#) Distance from $x=0$ to $x=\pm 1$ is 1

$\therefore$ $\boxed{\text{Radius of convergence} = 1}$

STEP-4 Assume a Power series solution and differentiate:

Theorem Existence of power series solutions:

If x_0 is an ordinary point of $y'' + P(x)y' + Q(x)y = 0$, the general solution has the form; $y = \sum_{n=0}^{\infty} c_n (x - x_0)^n$, and the series converges for $|x - x_0| < R$, where is the distance from x_0 to the nearest singular point.

[Refer step-3., where R → Radius of convergence.]

$$\left[y = \sum_{k=0}^{\infty} c_k x^k \right] \left[y' = \sum_{k=1}^{\infty} k c_k x^{k-1} \right] \left[y'' = \sum_{k=2}^{\infty} (k)(k-1) c_k x^{k-2} \right]$$

STEP-5 Substitute into each term of the ODE:

$$\underbrace{(1-x^2)y''}_{①} - \underbrace{2xy'}_{②} + \underbrace{2y}_{③} = 0$$

① $(1-x^2)y'' = y'' - x^2 y''$

$$= \sum_{k=2}^{\infty} (k)(k-1) c_k x^{k-2} - (x^2) \sum_{k=2}^{\infty} (k)(k-1) c_k x^{k-2}$$

$$= \sum_{k=2}^{\infty} (k)(k-1) c_k x^{k-2} - \sum_{k=2}^{\infty} (k)(k-1) c_k x^{k}$$

(2) $-2xy' = -2x\sum_{k=1}^{\infty} k c_k x^{k-1}$

$= -\sum_{k=1}^{\infty} 2k c_k x^k$

(3) $2y = 2\sum_{k=0}^{\infty} c_k x^k$

$= \sum_{k=0}^{\infty} 2c_k x^k$

STEP-6 Shift summation indices so that each term contains x^k:

(1) $y'' - x^2 y'' = \sum_{k=2}^{\infty} k(k-1) c_k x^{k-2}$

$- \sum_{k=2}^{\infty} k(k-1) c_k x^k$

$= \sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$

$- \sum_{k=0}^{\infty} (k)(k-1) c_k x^k$

$(k)(k-1) c_k = 0$ for $k = 0, 1$

(2) & (3) have x^k in them!

STEP-7 Combine terms ①, ② & ③ into one sum:

$$\sum_{n=0}^{\infty} \left[(k+2)(k+1)c_{k+2}\right] - \left[(k)(k-1)c_k\right] - \left[2kc_k\right] + \left[2c_k\right] x^k = 0$$

STEP-8 Set each coefficient to zero & derive the recurrence relation:

$$(k+2)(k+1)c_{k+2} = \left[(k)(k-1)c_k\right] + \left[2kc_k\right] - \left[2c_k\right]$$

$$= \left[(k)(k-1) + (2k) - 2\right] c_k$$

$$= (k^2 + k - 2)\, c_k$$

$$= (k+2)(k-1)\, c_k$$

$$\Rightarrow \quad c_{k+2} = \frac{(k+2)(k-1)c_k}{(k+2)(k+1)}$$

$$\Rightarrow \quad \boxed{c_{k+2} = \left(\frac{k-1}{k+1}\right) c_k \quad \text{where } k = 0, 1, 2, \ldots}$$

STEP-9 Generate the coefficients:

(#) C_0 & C_1 are arbitrary constants of the second order ODE

$$C_{k+2} = \left(\frac{k-1}{k+1}\right) C_k$$

Odd coefficients:

$k=1$: $C_3 = \frac{0}{2} C_1 = 0 \Rightarrow \boxed{C_3 = 0}$

$k=3$: $C_5 = \frac{2}{4} C_3 \Rightarrow \boxed{C_5 = 0}$

$k=5$: $C_7 = \frac{4}{6} C_5 \Rightarrow \boxed{C_7 = 0}$

$\therefore$ $y_2(x) = \sum_{k=0}^{\infty} C_k x^k$

for odd coefficients; $k = 1, 3, 5, \ldots$

$y_2(x) = C_1 x + \cancel{C_3 x^3}^{0} + \cancel{C_5 x^5}^{0} + \ldots$

$\Rightarrow \boxed{y_2(x) = C_1 x}$ for odd coefficients

Even coefficients:

$k=0$: $C_2 = \frac{-1}{1} C_0 \Rightarrow \boxed{C_2 = -C_0}$

$k=2$: $C_4 = \frac{1}{3} C_2 \Rightarrow \boxed{C_4 = -\frac{C_0}{3}}$

$k=4$: $C_6 = \frac{3}{5} C_4 \Rightarrow$ $\boxed{C_6 = \frac{-C_0}{5}}$

$k=6$: $C_8 = \frac{5}{7} C_6 \Rightarrow$ $\boxed{C_8 = \frac{-C_0}{7}}$

$\therefore y(x) = \sum_{k=0}^{\infty} C_k x^k$

for all even coefficients; $k=0, 2, 4 \cdots$

$$\boxed{y_1(x) = C_0 - C_0 x^2 - \frac{C_0}{3} x^4 - \frac{C_0}{5} x^6 - \cdots \quad \text{for all even coefficients}}$$

STEP-10 Write the general solution:

Sum of $y(x)$ for odd & even coefficients

$$\boxed{y(x) = C_0 \left[1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \cdots\right] + C_1 x}$$

i.e., $y(x) = C_0 y_1(x) + C_1 y_2(x)$

$$\Rightarrow \boxed{y_1(x) = 1 - \sum_{k=1}^{\infty} \frac{x^{2k}}{2k-1} \quad \& \quad y_2(x) = x}$$